LECTURE 1.2 · SHIP AND OCEAN SYSTEMS DESIGN

Owner’s Requirements

Papanikolaou (2026) §1.3.1, §1.3.5; §1.3.6.1–1.3.6.2 for Exercise 1.2 · Session W1.2, first half · Lecture 20 min + Exercise 1.2 20 min

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The Holy Bible, Genesis 6–8: Owner’s Requirements to Noah

Verse	The owner’s words (King James Version, 1611)	As a design requirement
6:14	Make thee an ark of gopher wood; rooms shalt thou make in the ark, and shalt pitch it within and without with pitch.	Timber hull, divided into compartments, sealed with pitch inside and outside
6:15	The length of the ark shall be three hundred cubits, the breadth of it fifty cubits, and the height of it thirty cubits.	L 135 m · B 22.5 m · D 13.5 m, with 1 cubit ≈ 0.45 m; L/B = 6, L/D = 10
7:20	Fifteen cubits upward did the waters prevail; and the mountains were covered.	Draught not stated; one reading: T = 15 cubits ≈ 6.75 m, half of D (an interpretation to max draught)
6:16	A window shalt thou make to the ark … and the door of the ark shalt thou set in the side thereof; with lower, second, and third stories shalt thou make it.	Light and air opening under the roof; one side door; three decks
6:18	… thou shalt come into the ark, thou, and thy sons, and thy wife, and thy sons' wives with thee.	8 persons on board, who are also the crew
6:19	… two of every sort shalt thou bring into the ark, to keep them alive with thee; they shall be male and female.	Cargo: live animals in pairs, many species: pens, feeding, handling
6:21	And take thou unto thee of all food that is eaten … and it shall be for food for thee, and for them.	Food for people and animals for the whole voyage; no resupply
7:12, 7:24	And the rain was upon the earth forty days and forty nights. … And the waters prevailed upon the earth an hundred and fifty days.	Watertight against 40 days of rain and months afloat
7:11, 8:14	In the six hundredth year of Noah's life, in the second month, the seventeenth day … And in the second month, on the seven and twentieth day … was the earth dried.	Endurance: just over one year (about 370 days), no port calls
7:18, 8:4	… the ark went upon the face of the waters. … And the ark rested … upon the mountains of Ararat.	No propulsion, no steering: it drifts, and must survive grounding

Potential solutions can be found in: https://www.fulltable.com/VTS/n/nk/na.htm

From an Owner’s Need to a Design Basis

1

The need

Why an owner orders a ship, and the mission it must serve

2

The requirements

What the owner specifies, and what is left to the designer

E

Exercise 1.2

Check that the owner’s brief adds up, and classify the ship, on the reference ship

Read: Papanikolaou (2026) §1.3.1, §1.3.5; §1.3.6.1–1.3.6.2 for Exercise 1.2

Bring: the reference ship’s owner’s brief (0.6, slide 2)

It Starts with a Business Need

Why owners order new ships

1. Replace aged or uncompetitive ships, or ships that no longer meet SOLAS or MARPOL
2. Extend or change a service already run
3. Open services in new areas
4. Carry new cargo on an existing line
5. Bring in new technology
6. Build special ships: tugs, offshore, research
7. Build floating offshore structures

How the need becomes a mission

1. Study the market and the fleet it needs
2. Compare: move an own ship, charter, buy second-hand or build new
3. If new: write the mission requirements, namely type and service, area, capacity, speed, range, crew, propulsion

> **Our two ships:** both replace an **ageing fleet** on the Azores lines (driver 1).

Every driver comes back to the **return on investment**. The mission requirements are the output of the owner’s decision, and the input of the design.

What the Owner Specifies

a · Capacity

Deadweight · Hold volume · TEU · Vehicles or lane metres · Passengers

b · Speed

Trial speed at the design draught, at 100% MCR or another agreed rating

c · Range

Sea miles or days without bunkering, at service speed; sizes fuel and other tanks

d · Class

A recognized classification society, chosen by the owner

Often added by the owner

Number of propellers · Engine type and maker, but not its power · Cargo handling system · Quality of accommodation

Usually left to the designer

Main dimensions, except canal, channel and port limits · Stability, except a minimum GM for some types

EXERCISE 1.2 · INDIVIDUAL · 20 MIN · PRACTICE FOR D1

Does the Owner’s Brief Add Up?

On **the reference ship**: its owner’s brief is 0.6, slide 2. Take the fuel at sea as about **25 t/day**, and water, stores and crew as about **300 t**.

1 Volume · 4 min

Does 35,000 t of cargo at 0.90 t/m³ fit in the cargo tanks at 98 % filling?

2 Deadweight · 5 min

37,600 t of deadweight minus 35,000 t of cargo: is what is left enough for fuel for the 10,000 nm range at 14.5 kn, plus water, stores and crew?

3 Other cargoes · 4 min

How much can the ship load of a light product at 0.75 t/m³, and of a heavy chemical at 1.20 t/m³? What runs out first, the tanks or the deadweight?

4 Classify · 2 min

m³ of cargo tank per t of deadweight: a deadweight or a volume carrier (Ship types 1.2)?

Check: the worked answer is shown at the end (Exercise Answers – Week 1). Not graded.

Your project: your group checks its own ship’s brief the same way outside class, for D1 (see your Project Scope Sheet).

The Story in Two Lines

> **1 · Need.** A business need, such as an ageing fleet, becomes a mission; the mission becomes the owner’s requirements.

> **2 · Requirements.** Capacity, speed, range and class; the main dimensions are usually left to the designer.

Next, in this session: Lecture 1.3, The Design Process. Keep your numbers from Exercise 1.2.